

# Swayam Jain

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## PROJECTS

### Quant Trading App | [GitHub](#) | [Live Demo](#)

June 2026

React, TypeScript, Python, FastAPI, PostgreSQL (pgvector / Supabase), LangGraph, Gemini API, SQLAlchemy, Pandas, NumPy

- **Architected an episodic strategy memory RAG pipeline** utilizing Supabase pgvector (HNSW cosine search) and Gemini 768-dim embeddings; classified market conditions into 6 regimes to retrieve analogous trade episodes and flag false breakout traps (<40% win rate).
- **Built an autonomous LLM research agent** via LangGraph (ReAct framework) orchestrating 6 specialized financial tools (pricing, technicals, peer momentum, news sentiment, backtesting, vector memory) to produce cited BUY/HOLD/SELL trade theses in ~30s.
- **Engineered a parallel stock-screening engine** processing 134 NSE-listed equities via a multi-threaded worker pool, computing real-time SMA, RSI, and ATR technical indicators to rank momentum candidates in under 2 seconds per scan pass.
- **Developed a multi-tenant FastAPI backend** with Supabase PostgreSQL, JWT auth, and row-level data isolation; integrated an automated 3% stop-loss liquidation engine and a historical backtesting pipeline (Sharpe ratio, max drawdown, win rate).

### Expense Tracker via WhatsApp | [GitHub](#)

August 2026

TypeScript, React, Node.js, Meta Cloud API, Gemini AI, PostgreSQL (Supabase), Vercel

- **Built a conversational financial tracking agent** integrating Meta Cloud API (WhatsApp Webhooks) and Gemini AI to parse unstructured natural-language inputs (e.g., "rs 247 debit food") and execute zero-friction transaction logging.
- **Designed a multi-bucket balance tracking architecture** (bank accounts, liquid cash, custom funds) featuring auto-resetting monthly budget thresholds and an automated start-of-month salary-confirmation reconciliation workflow.
- **Developed a full-stack financial dashboard** in React and TypeScript with a Node.js/Supabase backend, supporting real-time transaction auditing, bucket lifecycle operations (add/edit/delete), and historical debit/credit visualizations.

### Stock Trading Simulation Game | [GitHub](#)

April 2026

Python, FastAPI, React, TypeScript, Vite, Material UI, yfinance

- **Built an Indian equity (NSE) paper-trading platform** integrating real-time market data streams via yfinance, allowing users to execute simulated limit/market orders with virtual currency and zero capital exposure.
- **Engineered a client-side portfolio management engine** rendering interactive line and candlestick charts to compute dynamic asset allocation, total position valuation, and real-time unrealized/realized P&L calculations.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, TypeScript, JavaScript

**Frameworks & Libraries:** FastAPI, React, Node.js, LangGraph, LangChain (RAG), SQLAlchemy, Pandas, NumPy, Scikit-Learn, Matplotlib, Vite, Material UI

**Databases:** PostgreSQL (pgvector), Supabase, MySQL, SQLite, MongoDB

**APIs & Cloud:** REST APIs, Google Gemini API, Meta Cloud API, Vercel

**Developer Tools:** Git, GitHub, Power BI, Tableau

## EDUCATION

### SRM Institute of Science and Technology - Ghaziabad

2023 - 2027

Bachelor of Technology (B.Tech) in Computer Science and Engineering

CGPA: 8.17 / 10.0

## CERTIFICATIONS

### Tata Group - Data Analytics Job Simulation (Tata iQ) | [Certificate](#)

August 2026

- Conducted exploratory data analysis (EDA) and credit risk indicator modeling; architected an agentic AI delinquency recovery and collections strategy.

### NPTEL - Natural Language Processing (IIT) | [Certificate](#)

April 2026

- Completed rigorous curriculum covering tokenization, statistical language models, sequence-to-sequence networks, and transformer-based NLP pipelines.

## RELEVANT COURSEWORK

Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Object-Oriented Design & Programming (OOP), Computer Networks, Artificial Intelligence, Data Science, Data Mining & Analytics

## EXTRACURRICULAR ACTIVITIES

- **Competitive Programming & Technical Exploration:** Independently research quantitative trading strategies and financial markets beyond the academic curriculum.
- **Strategic Gaming & Esports:** Compete in multiplayer and strategy-based games requiring analytical thinking and fast decision-making under pressure.